



Guangdong Technion

Israel Institute of Technology

广东以色列理工学院

GTIIT

CALCULUS 1 - M
WINTER SEMESTER - 2024

LECTURE 13 - NOVEMBER 25TH
SECOND PART

ANTIDERIVATIVES

Position is the Antiderivative of the Velocity as the Velocity is the derivative of the Position

A biologist who knows the **rate** $f'(t)$ at which a bacteria population is **increasing** might want to deduce what the size $f(t)$ of the population will be at some future time.

The problem is to find a function F **whose derivative is a known function** f . If such a function F exists, it is called an antiderivative of f :

$$F(x) = \int f(x) dx$$

The operation is called **integration**.

$$F(x) = \int f(x)dx$$

Definition. A function F is called an antiderivative of f on an open interval (a, b) if

$$F'(x) = f(x) \quad \forall x \in (a, b)$$

- If F is an antiderivative of f on (a, b) , then, $\forall C \in \mathbb{R}$:

$$F(x) + C \quad \text{is an antiderivative of } f \text{ on } (a, b)$$

- Notation: any antiderivative is denoted by

$$F(x) = \int f(x)dx$$

All the Antiderivatives of a given f are $F(x) + C$

Example 1. Find all the Antiderivatives of $f(x) = x^2$.

Solution. We know that $\frac{d}{dx} [x^3] = 3x^2$, then

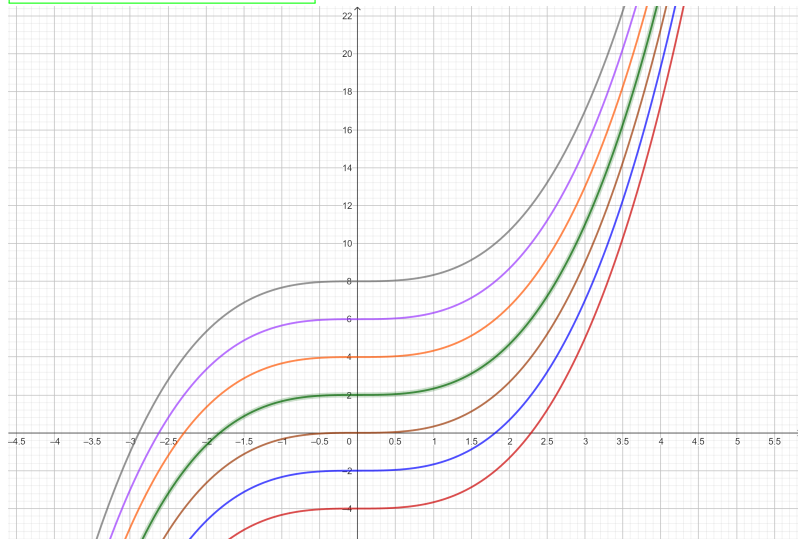
$$\frac{d}{dx} \left[\frac{1}{3} x^3 \right] = x^2$$

Hence, all the antiderivatives of $f(x) = x^2$ on $\mathbb{R}$ are

$$F(x) = \int x^2 dx = \frac{1}{3} x^3 + C : C \in \mathbb{R}$$

All the Antiderivatives of a given f are $F(x) + C$

$$F(x) = \int x^2 dx = \frac{1}{3}x^3 + C : C \in \mathbb{R}$$



All the Antiderivatives of a given f are $F(x) + C$

Example 2. Find all the Antiderivatives of $f(x) = \frac{1}{x}$.

Solution. We know that $\frac{d}{dx} [\ln(x)] = \frac{1}{x}$ for $x > 0$, and also, $|x| = -x$ if $x < 0$, so

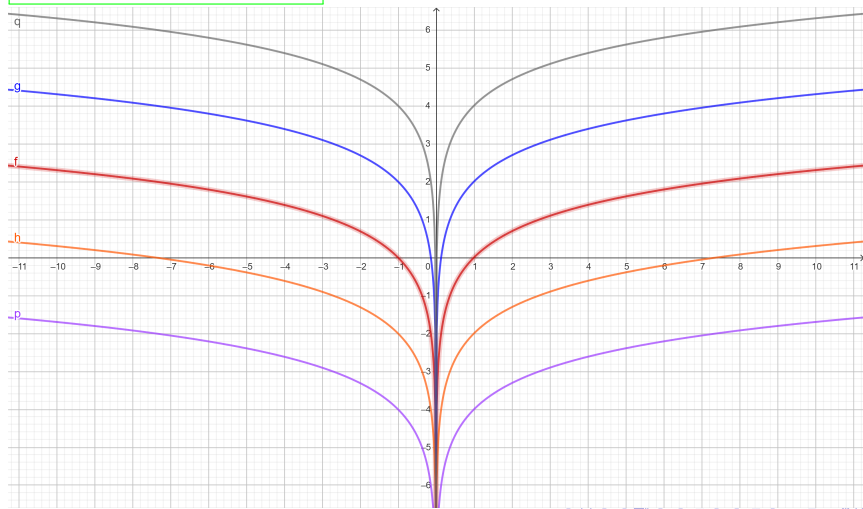
$$\frac{d}{dx} [\ln(-x)] = \frac{-1}{-x} = \frac{1}{x} \text{ for } x < 0$$

Then all the antiderivatives of $f(x) = \frac{1}{x}$ on $(-\infty, 0) \cup (0, \infty)$ are

$$F(x) = \int \frac{1}{x} dx = \ln |x| + C : C \in \mathbb{R}$$

Antiderivatives $\int \frac{1}{x} dx = \ln |x| + C$

$$F(x) = \int \frac{1}{x} dx = \ln |x| + C : C \in \mathbb{R}$$



Rules of Integration

Let F, G be antiderivatives of f and g , respectively, then $F + G$ is an antiderivative of $f + g$:

$$(F + G)(x) = \int f(x)dx + \int g(x)dx = \int (f(x) + g(x))dx$$

If $c \in \mathbb{R}$, then

$$(c \cdot F)(x) = c \cdot \int f(x)dx = \int c \cdot f(x)dx$$

Rules of Integration

Example 3. Find all the Antiderivatives of $f(x) = x^n$.

Solution. We know that $\frac{d}{dx} [x^{n+1}] = (n+1)x^n$, then

$$\frac{d}{dx} \left[\frac{1}{(n+1)} x^{n+1} \right] = x^n$$

Hence, all the antiderivatives of $f(x) = x^n$ on $\mathbb{R}$ are

$$F(x) = \int x^n dx = \frac{1}{(n+1)} x^{n+1} + C : C \in \mathbb{R}$$

Rules of Integration

Some particular examples are

$$\int 1 \, dx = x + C,$$

$$\int x \, dx = \frac{1}{2}x^2 + C,$$

$$\int x^{1/2} \, dx = \frac{2}{3}x^{3/2} + C,$$

$$\int x^3 \, dx = \frac{1}{4}x^4 + C :$$

$$\int x^{1/3} \, dx = \frac{3}{4}x^{4/3} + C : C \in \mathbb{R}$$

Use the Rules to find the integral

a)

$$\int (\pi x + 2x^{3/5} - 1) dx$$

$$b) \int (2x^{2/5} - 6x^2 + 8) dx$$

Solution. a)

$$\int (\pi x + 2x^{3/5}) dx = \pi \int x dx + 2 \int x^{3/5} dx - \int 1 dx$$

$$= \pi \cdot \frac{1}{2} x^2 + 2 \cdot \frac{5}{8} x^{8/5} - x + C : C \in \mathbb{R} \checkmark$$

$$b) \int (2x^{2/5} - 6x^2 + 8) dx = 2 \int x^{2/5} dx - 6 \int x^2 dx + 8 \int 1 dx$$

$$= 2 \cdot \frac{5}{7} x^{7/5} - 2x^3 + 8x + C : C \in \mathbb{R} \checkmark$$

Rules of Integration of the Main Functions

The following are the **antiderivatives** of the functions on the left, in their respective domains:

$f(x)$	$F(x) = \int f(x)dx$
0	C
1	x
x^n	$\frac{1}{n+1}x^{n+1}$
$\sqrt{x}$	$\frac{2}{3}x^{3/2}$ only for $x > 0$
x^r	$\frac{1}{r+1}x^{r+1}$, $r \neq -1$, only for $x \in \text{Dom}(f)$, $x \neq 0$
$\frac{1}{x}$	$\ln x $
e^x	e^x
$\sin x$	$-\cos x$
$\cos x$	$\sin x$
$\frac{1}{\cos^2 x}$	$\tan x$

Rules of Integration of the Main Functions

$$F(x) = \int f(x)dx$$

$$\int 0 \, dx = C, \quad \int 1 \, dx = x$$

$$\int \sqrt{x} \, dx = \frac{2}{3}x^{3/2} \text{ only for } x > 0$$

$$\int x^r \, dx = \frac{1}{r+1}x^{r+1}, r \neq -1, \text{ only for } x \in \text{Dom}(f), x \neq 0$$

$$\int \frac{1}{x} \, dx = \ln |x|$$

$$\int e^x dx = e^x, \quad \int a^x dx = \frac{1}{\ln(a)}a^x$$

$$\int \sin x \, dx = -\cos x, \quad \int \cos x \, dx = \sin x$$

$$\int \frac{1}{\cos^2 x} dx = \tan x$$

Rules of Integration of the Main Functions

$$F(x) = \int f(x) dx$$

$$\int \cosh(x) dx = \sinh(x)$$

$$\int \sinh(x) dx = \cosh(x)$$

$$\int \frac{1}{1+x^2} dx = \tan^{-1}(x)$$

$$\int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1}(x)$$

$$\int \frac{-1}{\sqrt{1-x^2}} dx = \cos^{-1}(x)$$

Use the table of integrals to solve the differential equation

Find f such that $f(x) = \int \left[e^x + 17(1+x^2)^{-1} \right] dx$ and $f(0) = -4$

Solution. Let us compute f as the antiderivative:

$$\bullet f(x) = \int \left[e^x + 17(1+x^2)^{-1} \right] dx$$

$$= e^x + 17 \int \frac{1}{1+x^2} dx = \boxed{e^x + 17 \tan^{-1}(x) + C}$$

• Besides, f must verify $f(0) = -4$:

$$-4 = f(0) = \left[e^x + 17 \tan^{-1}(x) + C \right] \Big|_{x=0} = e^0 + 17 \tan^{-1}(0) + C$$

$$\boxed{= 1 + C \Rightarrow C = -5 \checkmark}$$

Hence, there is a unique solution

$$f(x) = e^x + 17 \tan^{-1}(x) - 5 \checkmark$$

Integration by Substitution

It is a consequence of the chain rule for derivatives

$$\int e^{\pi x} dx = \frac{1}{\pi} e^{\pi x} + C$$

$$\int e^{\sin(x)} \cos(x) dx = e^{\sin(x)} + C$$

$$\int \sin^2(x) \cos(x) dx = \frac{1}{3} \sin^3(x) + C$$

Integration by Substitution

It is a consequence of the chain rule for the derivative of a composition

Given

$$\int e^{\sin(x)} \cos(x) dx$$

consider the change of variables $u = \sin(x)$, so

$$\int e^{\sin(x)} \cos(x) dx$$

$$= \int e^{u(x)} u'(x) dx = \int e^{u(x)} \frac{du}{dx} dx = \int e^u du$$

$$= e^u = e^{\sin(x)} + C \checkmark$$

Integration by Substitution

Solve $\int \frac{\sin(x)}{\sqrt{\cos(x)}} dx$

Solution.

- Take $u = \cos(x) \Rightarrow du = u'(x) dx = -\sin(x) dx$

- $\int \frac{1}{\sqrt{u}} du = \int u^{-1/2} du$, and we solve it with $r = -1/2$,
then

$$\int \frac{1}{\sqrt{u}} du = \int u^r du = \frac{1}{r+1} u^{r+1} = \frac{1}{-\frac{1}{2}+1} u^{(-\frac{1}{2}+1)} = 2\sqrt{u} + C$$

$$\Rightarrow \int \frac{\sin(x)}{\sqrt{\cos(x)}} dx = \int \frac{(-du)}{\sqrt{u}} = - \int \frac{1}{\sqrt{u}} du = -2\sqrt{u} + C$$

$$= -2\sqrt{\cos(x)} + C \checkmark$$

Integration by Substitution

Solve $\int \frac{1}{4 + 3x^2} dx$

Solution.

- $\int \frac{1}{4 + 3x^2} dx = \int \frac{1}{4\left(1 + \frac{3x^2}{4}\right)} dx = \frac{1}{4} \int \frac{1}{\left[1 + \left(\frac{\sqrt{3}}{2}x\right)^2\right]} dx$

- Take $u = \frac{\sqrt{3}}{2}x \Rightarrow du = u'(x) dx = \frac{\sqrt{3}}{2}dx$, then $\frac{2}{\sqrt{3}}du = dx$, so

$$\int \frac{1}{4 + 3x^2} dx = \frac{1}{4} \int \frac{1}{\left[1 + \left(\frac{\sqrt{3}}{2}x\right)^2\right]} dx = \frac{2}{4\sqrt{3}} \int \frac{1}{[1 + u^2]} du$$

$$= \frac{1}{2\sqrt{3}} \tan^{-1}(u) + C = \frac{1}{2\sqrt{3}} \tan^{-1}\left(\frac{\sqrt{3}}{2}x\right) + C \checkmark$$